

Qualification Pack



Sugar Processing Operator

Electives: Clarification of sugarcane juice in a sugar mill/ Perform evaporation of sugarcane Juice in a sugar mill/ Perform crystallization of sugar syrup in a sugar mill/ Perform refining of crystal sugar/ Perform drying of crystal sugar-

QP Code: FIC/Q7102

Version: 1.0

NSQF Level: 4

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Qualification Pack

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FIC/Q7102: Sugar Processing Operator

Brief Job Description

Sugar processing Operator is in charge of ensuring that machinery and equipment needed to produce sugar from raw materials like sugar cane or sugar beets are operated efficiently and safely. They control and direct the whole procedure, from the extraction of sugarcane juice to the production of refined sugar products. Sugar processing operator make sure that the product is produced in accordance with industry standards for quality, safety, and the environment.

Personal Attributes

They must have unrelenting attention to detail, to uphold the highest levels of quality control. Prioritizing and fostering a culture of safety among their team, they place a high priority on safety consciousness. They can deal with the constantly shifting requirements of sugar processing because of their high problem-solving abilities and adaptability. For seamless cooperation, effective communication and teamwork are required, and stress management enables serenity under pressure. The critical personality qualities necessary for success in this profession also include dependability, moral behaviour, and a dedication to creating premium sugar goods.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [FIC/N9026: Prepare for production](#)
2. [FIC/N7103: Carry out sugar processing](#)
3. [FIC/N7104: Carry out Maintenance and Troubleshooting](#)
4. [FIC/N9906: Apply food safety guidelines in Food Processing](#)
5. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Electives(*mandatory to select at least one*):

Elective 1: Clarification of sugarcane juice in a sugar mill

This OS unit is about removing the impurities from the sugarcane juice obtained from the milling process as per the industry standard and the available machinery.

1. [FIC/N7105: Perform clarification of sugarcane juice in a sugar mill](#)

Elective 2: Perform evaporation of sugarcane Juice in a sugar mill

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This OS unit is about overseeing the evaporation stage of sugar production in order to get a concentrated sugarcane juice for extracting sugar crystals as per the industry standard and the available machinery.

1. [FIC/N7106: Perform Evaporation of sugarcane Juice in a sugar mill](#)

Elective 3: Perform crystallization of sugar syrup in a sugar mill

This OS unit is about overseeing the crystallization process in which the concentrated sugar syrup is converted into sugar crystals as per the industry standard and the available machineries.

1. [FIC/N7107: Perform crystallization of sugarcane Juice in a sugar mill](#)

Elective 4: Perform refining of crystal sugar

This OS unit is about managing refining process of the sugar crystals as per the industry standard and the available machinery.

1. [FIC/N7108: Perform refining of crystal sugar](#)

Elective 5: Perform drying of crystal sugar-

This OS unit is about removing the moisture from the sugar crystals in the drying section and ensure the final product meets the quality standards. They are also responsible for the maintenance and troubleshooting of the equipment used in the process.

1. [FIC/N7109: Perform drying of crystal sugar](#)

Qualification Pack (QP) Parameters

Sector	Food Processing
Sub-Sector	Confectionery
Occupation	Processing-Confectionery
Country	India
NSQF Level	4
Credits	41
Aligned to NCO/ISCO/ISIC Code	NCO-2015/2141.1800, NCO-2015/8160.3700, NCO-2015/8160.3800, NCO-2015/8160.3900, NCO-2015/8160.5300, NCO-2015/2131.1400

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Minimum Educational Qualification & Experience	12th grade Pass OR Completed 2nd year of 3-year diploma (after 10th) and pursuing regular diploma OR 10th grade pass with 3 Years of experience *Relevant experience in the Food/Sugar processing industry OR Previous relevant Qualification of NSQF Level (3) with 3 Years of experience *Relevant experience in the Food/Sugar processing industry
Minimum Level of Education for Training in School	Ability to read and write
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	29/05/2027
NSQC Approval Date	30/05/2024
Version	1.0
Reference code on NQR	QG-04-FI-02617-2024-V1-FICSI
NQR Version	1.0

Remarks:

NA

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FIC/N9026: Prepare for production

Description

This NOS unit is about performing various tasks prior to production in the food processing industry.

Scope

The scope covers the following :

- Plan for production
- Clean and maintain work area, machineries, and tools for production
- Organize for production

Elements and Performance Criteria

Plan for production

To be competent, the user/individual on the job must be able to:

- PC1.** identify work requirements by obtaining instructions from the supervisor. Instructions: process chart, product flow chart, formulation, chart, etc.
- PC2.** plan and prioritize tasks as per work schedule. Tasks: inspect, clean, maintain, verify, etc.
- PC3.** estimate manpower and material requirements as per work requirement. Material: raw materials and packaging materials
- PC4.** ensure required quantity of raw materials, packaging materials, equipment, and manpower for production
- PC5.** plan capacity utilization of machinery with respect to the processing time, production order, and batch size for each product

Clean and maintain work area, machineries, and tools for production

To be competent, the user/individual on the job must be able to:

- PC6.** clean and maintain the work area as per organizational procedures
- PC7.** clean and maintain the machines and tools and sanitize them as per the organization's specifications and standards
- PC8.** dispose of the waste material at designated place safely. Waste material: hazardous waste, food waste, packaging waste, etc.
- PC9.** inspect the tools, equipment, and machinery to ascertain suitability for use
- PC10.** report information such as faulty tools and equipment to the concerned authority

Organize for production

To be competent, the user/individual on the job must be able to:

- PC11.** organize tools and equipment
- PC12.** receive and organize production materials appropriately. Production materials: raw materials, packaging materials, etc.
- PC13.** allot responsibilities/work to the assistants and helpers

Knowledge and Understanding (KU)

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The individual on the job needs to know and understand:

- KU1.** production planning process
- KU2.** analysis and interpretation of various process charts, product flow charts, etc.
- KU3.** resource management process
- KU4.** procedure to estimate manpower and raw material
- KU5.** capacity utilization calculation
- KU6.** organizational policies and SOP on cleanliness
- KU7.** operating procedure and general maintenance of food production machineries
- KU8.** waste management procedures
- KU9.** methods to inspect tools, equipment and machinery
- KU10.** procedure to allot work or responsibility to the team

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret organizational policies, SOP, production charts, etc.
- GS2.** communicate effectively with subordinates as well as supervisors
- GS3.** plan and prioritize various tasks
- GS4.** be always punctual and courteous
- GS5.** organize all process/equipment manuals to access information easily
- GS6.** discuss task lists, schedules, and activities with the senior/supervisor

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Plan for production</i>	11	25	-	-
PC1. identify work requirements by obtaining instructions from the supervisor. Instructions: process chart, product flow chart, formulation, chart, etc.	3	6	-	-
PC2. plan and prioritize tasks as per work schedule.Tasks: inspect, clean, maintain, verify, etc.	2	5	-	-
PC3. estimate manpower and material requirements as per work requirement. Material: raw materials and packaging materials	2	4	-	-
PC4. ensure required quantity of raw materials, packaging materials, equipment, and manpower for production	2	5	-	-
PC5. plan capacity utilization of machinery with respect to the processing time, production order, and batch size for each product	2	5	-	-
<i>Clean and maintain work area, machineries, and tools for production</i>	14	32	-	-
PC6. clean and maintain the work area as per organizational procedures	3	7	-	-
PC7. clean and maintain the machines and tools and sanitize them as per the organization's specifications and standards	3	7	-	-
PC8. dispose of the waste material at designated place safely. Waste material: hazardous waste, food waste, packaging waste, etc.	3	7	-	-
PC9. inspect the tools, equipment, and machinery to ascertain suitability for use	3	6	-	-
PC10. report information such as faulty tools and equipment to the concerned authority	2	5	-	-
<i>Organize for production</i>	5	13	-	-
PC11. organize tools and equipment	2	7	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. receive and organize production materials appropriately. Production materials: raw materials, packaging materials, etc.	2	4	-	-
PC13. allot responsibilities/work to the assistants and helpers	1	2	-	-
NOS Total	30	70	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	FIC/N9026
NOS Name	Prepare for production
Sector	Food Processing
Sub-Sector	Generic
Occupation	Production
NSQF Level	3
Credits	TBD
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2029
NSQC Clearance Date	06/02/2026

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FIC/N7103: Carry out sugar processing

Description

This NOS unit is about sugar processing using various methods and machineries as per the specification and standards of the organization.

Scope

The scope covers the following :

- Pre-processing of Resources
- Processing of Sugar
- Conduct Quality Tests and Inspections
- Carry out Packaging and Dispatch

Elements and Performance Criteria

Pre-Processing of Sugar

To be competent, the user/individual on the job must be able to:

- PC1.** check safety norms of the defined area and inform the safety in charge if not suitable
- PC2.** interpret the process flow chart/formulation chart for products to be produced
- PC3.** inspect the quantity and quality of the received raw material against specification to ensure it meets the quality standards. Raw material: Sugar cane etc. Quality standards: Purity and cleanliness, Chemical Composition, Pesticide and chemical residue levels, etc.
- PC4.** maintain inventory while effectively storing raw material in designated areas
- PC5.** prepare and set up the machines for sugar processing. Machines: Extraction machinery, Cane preparatory devices, Mills, etc.
- PC6.** conduct equipment inspection and maintenance to ensure good operator order
- PC7.**
 - prepare quality control tools and devices for monitoring essential factors.
 - Factors: Humidity, temperature, pressure, pH, etc.
- PC8.** calibrate and validate the quality control equipment for accurate test results
- PC9.**
 - ensure there are ample supplies of packaging materials for packing the finished sugar product.
 - Packaging materials: cartons, Sugar bags, containers, etc.
- PC10.** check the cleanliness of packaging material to ensure it fulfils hygiene and quality requirements
- PC11.**
 - perform safety inspections and ensure that all safety equipment is accessible and in good working order.
 - Equipment: personal equipment, emergency response equipment, etc.

Processing of Sugar

To be competent, the user/individual on the job must be able to:

- PC12.** manage sugar extraction equipment, such as cane preparatory devices, mills, etc. to extract the juice from sugarcane

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- PC13.** • oversee and update the extraction parameters to maximize juice extraction and sugar yield
• Parameters: crushing or milling speed, Pressure, Feed rate, cutting size, extraction time, temperature, moisture content, sucrose content, etc.
- PC14.** clean sugarcane juice from contaminants and solids by following the clarification process
- PC15.** remove suspended particles by using filtering machinery to ensure juice clarity
- PC16.** control the evaporation process to lower the water content and raise sugar concentration
- PC17.** • monitor and adjust parameters of the evaporation equipment.
• Parameters: temperature, pressure, flow rates, etc.
- PC18.** manage crystallization process for creating sugar crystals
- PC19.** • alter variables to get desired size and quality of crystals.
• Variables: seeding, temperatures, etc.
- PC20.** separate sugar crystals from molasses-containing liquid through centrifuges
- PC21.** administer drying process to completely dry sugar crystals
- PC22.** sustain ideal conditions and temperatures to ensure product quality

Conduct quality tests and inspections

To be competent, the user/individual on the job must be able to:

- PC23.** • monitor the efficiency of crushers or mills by observing essential factors while extracting sugar cane juice.
• Factors: juice flow rates, pulp discharge, etc.
- PC24.** • examine extracted juice to assess its initial parameters.
• Parameters: sugar concentration, colour, clarity, etc.
- PC25.** administer the clarifying and filtering operations to confirm the removal of contaminants and suspended particles
- PC26.** conduct routine turbidity and suspended solids testing to evaluate the working of filtration and clarifying process
- PC27.** oversee the evaporation and concentration processes to decrease water content and raise sugar concentration
- PC28.** regulate brix (sugar concentration) testing to ensure it complies with specified requirements
- PC29.** analyze the crystal samples periodically to verify that size and purity are within standards
- PC30.** analyze sugar recovery rate and separation efficiency
- PC31.** check product's temperature and moisture content on a regular basis
- PC32.** oversee extensive quality checks on purity, color, sucrose content and particle size

Carry out packaging and dispatch

To be competent, the user/individual on the job must be able to:

- PC33.** keep packaged materials dry and spotless to avoid contamination or damage
- PC34.** operate filling machines to precisely weigh and fill packing containers with required amount of sugar
- PC35.** examine filling procedure to avoid overfilling or underfilling
- PC36.** • apply appropriate details of the product by employing labelling and coding machines.
• Details: product labels, barcodes, lot numbers, expiration dates, etc.
- PC37.** check packaged goods on a regular basis for any flaws. Flaws: contaminated or broken packing
- PC38.** remove and separate goods that are defective or broken

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- PC39.** organize stacking and palletization of packed goods for effective storage and transportation
- PC40.** keep thorough records of the packing procedure and revisions made to the package.
Records: including the number of items packed, the packaging materials employed, any errors or deviation etc
- PC41.** plan and manage loading of packaged sugar goods effectively and safely onto vehicles or shipping containers
- PC42.** • verify specifics of packaged sugar goods and shipment or delivery correspond to customer's demands.
• Specifics of packaging: amount, kind of product, packaging requirement, etc.
• Specifics of shipment and delivery: final destination, time of delivery, etc.
- PC43.** • communicate periodically with teams to organize delivery schedules.
• Teams: Logistic team, dispatchers, transport staff, etc.
- PC44.** resolve any packing or dispatch-related problems and notify to supervisor

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** fundamental principles behind the process of sugar extraction, crystallization, and purification
- KU2.** physical and chemical characteristics of sugar and its behaviour when processed
- KU3.** how to operate and maintain sugar processing apparatus, including mills, centrifuges, evaporators, and packaging equipment
- KU4.** characteristics and features of each processing line machine
- KU5.** sugar product quality requirements including sucrose concentration, colour, purity, and particle size
- KU6.** how to do quality checks and evaluate test findings
- KU7.** safety procedures and guidelines in a manufacturing setting, including hazardous materials and mechanical safety
- KU8.** environmental legislation relating to sustainable practices and trash disposal
- KU9.** effects of process control variables on product quality, such as, temperature, pressure, flow rates, and sugar concentration
- KU10.** processing parameters to optimize efficiency and yield
- KU11.** good hygiene practices and food safety regulations
- KU12.** how to identify and address technical problems, quality-standard violations and equipment problems
- KU13.** inventory management and material handling
- KU14.** handling techniques for raw materials, packaging materials and finished products
- KU15.** how to maintain accurate records of processing activities, quality tests, equipment maintenance and production logs
- KU16.** industry-specific regulations, standards, and certifications related to sugar processing
- KU17.** compliance requirements for safety, quality, and environmental practices

Generic Skills (GS)

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User/individual on the job needs to know how to:

- GS1.** knowledge of documentation and record-keeping
- GS2.** analyze information and take appropriate actions
- GS3.** prioritize safety and adhering to safety protocols
- GS4.** note down the information communicated by the senior/supervisor, observations (if any) related to the process, as per applicability in the organization
- GS5.** read and interpret the work order from the supervisor, certificate of analysis, analysis report from internal lab, process required for producing various types of products, equipment manuals and process documents to understand the equipment operation and process requirement
- GS6.** communicate effectively with subordinates as well as supervisors
- GS7.** perform calculations related to quantities, parameters and measurements
- GS8.** collaborate and contribute to a positive and productive work environment
- GS9.** • discuss task lists, schedules, and activities for carrying out production with the senior/supervisor
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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Pre-Processing of Sugar</i>	6	11	-	9
PC1. check safety norms of the defined area and inform the safety in charge if not suitable	-	-	-	-
PC2. interpret the process flow chart/formulation chart for products to be produced	-	-	-	-
PC3. inspect the quantity and quality of the received raw material against specification to ensure it meets the quality standards. Raw material: Sugar cane etc. Quality standards: Purity and cleanliness, Chemical Composition, Pesticide and chemical residue levels, etc.	-	-	-	-
PC4. maintain inventory while effectively storing raw material in designated areas	-	-	-	-
PC5. prepare and set up the machines for sugar processing. Machines: Extraction machinery, Cane preparatory devices, Mills, etc.	-	-	-	-
PC6. conduct equipment inspection and maintenance to ensure good operator order	-	-	-	-
PC7. • prepare quality control tools and devices for monitoring essential factors. • Factors: Humidity, temperature, pressure, pH, etc.	-	-	-	-
PC8. calibrate and validate the quality control equipment for accurate test results	-	-	-	-
PC9. • ensure there are ample supplies of packaging materials for packing the finished sugar product. • Packaging materials: cartons, Sugar bags, containers, etc.	-	-	-	-
PC10. check the cleanliness of packaging material to ensure it fulfils hygiene and quality requirements	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. - perform safety inspections and ensure that all safety equipment is accessible and in good working order. - Equipment: personal equipment, emergency response equipment, etc.	-	-	-	-
<i>Processing of Sugar</i>	6	12	-	9
PC12. manage sugar extraction equipment, such as cane preparatory devices, mills, etc. to extract the in to juice from sugarcane	-	-	-	-
PC13. - oversee and update the extraction parameters to maximize juice extraction and sugar yield - Parameters: crushing or milling speed, Pressure, Feed rate, cutting size, extraction time, temperature, moisture content, sucrose content, etc.	-	-	-	-
PC14. clean sugarcane juice from contaminants and solids by following the clarification process	-	-	-	-
PC15. remove suspended particles by using filtering machinery to ensure juice clarity	-	-	-	-
PC16. control the evaporation process to lower the water content and raise sugar concentration	-	-	-	-
PC17. - monitor and adjust parameters of the evaporation equipment. - Parameters: temperature, pressure, flow rates, etc.	-	-	-	-
PC18. manage crystallization process for creating sugar crystals	-	-	-	-
PC19. - alter variables to get desired size and quality of crystals. - Variables: seeding, temperatures, etc.	-	-	-	-
PC20. separate sugar crystals from molasses-containing liquid through centrifuges	-	-	-	-
PC21. administer drying process to completely dry sugar crystals	-	-	-	-
PC22. sustain ideal conditions and temperatures to ensure product quality	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Conduct quality tests and inspections</i>	5	10	-	8
PC23. - monitor the efficiency of crushers or mills by observing essential factors while extracting sugar cane juice. - Factors: juice flow rates, pulp discharge, etc.	-	-	-	-
PC24. - examine extracted juice to assess its initial parameters. - Parameters: sugar concentration, colour, clarity, etc.	-	-	-	-
PC25. administer the clarifying and filtering operations to confirm the removal of contaminants and suspended particles	-	-	-	-
PC26. conduct routine turbidity and suspended solids testing to evaluate the working of filtration and clarifying process	-	-	-	-
PC27. oversee the evaporation and concentration processes to decrease water content and raise sugar concentration	-	-	-	-
PC28. regulate brix (sugar concentration) testing to ensure it complies with specified requirements	-	-	-	-
PC29. analyze the crystal samples periodically to verify that size and purity are within standards	-	-	-	-
PC30. analyze sugar recovery rate and separation efficiency	-	-	-	-
PC31. check product's temperature and moisture content on a regular basis	-	-	-	-
PC32. oversee extensive quality checks on purity, color, sucrose content and particle size	-	-	-	-
<i>Carry out packaging and dispatch</i>	6	12	-	6
PC33. keep packaged materials dry and spotless to avoid contamination or damage	-	-	-	-
PC34. operate filling machines to precisely weigh and fill packing containers with required amount of sugar	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC35. examine filling procedure to avoid overfilling or underfilling	-	-	-	-
PC36. • apply appropriate details of the product by employing labelling and coding machines. • Details: product labels, barcodes, lot numbers, expiration dates, etc.	-	-	-	-
PC37. check packaged goods on a regular basis for any flaws. Flaws: contaminated or broken packing	-	-	-	-
PC38. remove and separate goods that are defective or broken	-	-	-	-
PC39. organize stacking and palletization of packed goods for effective storage and transportation	-	-	-	-
PC40. keep thorough records of the packing procedure and revisions made to the package. Records: including the number of items packed, the packaging materials employed, any errors or deviation etc	-	-	-	-
PC41. plan and manage loading of packaged sugar goods effectively and safely onto vehicles or shipping containers	-	-	-	-
PC42. • verify specifics of packaged sugar goods and shipment or delivery correspond to customer's demands. • Specifics of packaging: amount, kind of product, packaging requirement, etc. • Specifics of shipment and delivery: final destination, time of delivery, etc.	-	-	-	-
PC43. • communicate periodically with teams to organize delivery schedules. • Teams: Logistic team, dispatchers, transport staff, etc.	-	-	-	-
PC44. resolve any packing or dispatch-related problems and notify to supervisor	-	-	-	-
NOS Total	23	45	-	32

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National Occupational Standards (NOS) Parameters

NOS Code	FIC/N7103
NOS Name	Carry out sugar processing
Sector	Food Processing
Sub-Sector	
Occupation	Processing-Confectionery
NSQF Level	4
Credits	4
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	29/05/2027
NSQF Clearance Date	30/05/2024

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FIC/N7104: Carry out Maintenance and Troubleshooting

Description

This NOS unit is about carrying out preventive maintenance and troubleshooting of equipment issues along with periodic documentation.

Scope

The scope covers the following :

- The scope covers the following:
- Carry out preventive planned maintenance
- Perform troubleshooting of equipment issues
- Carry out documentation to ensure compliance

Elements and Performance Criteria

Carry out preventive planned maintenance

To be competent, the user/individual on the job must be able to:

- PC1.**
 - check for wear, damage, or malfunction on a regular basis on all processing machinery.
 - Machinery: mills, juice heaters, clarifier, evaporators, pans, crystallisers, centrifuge, , drying units, etc.
- PC2.** perform operational and visual inspections to identify any anomalies or possible problems
- PC3.** lubricate appropriately moving components to decrease friction and prevent wear
- PC4.**
 - prevent performance issues by keeping equipment away from contaminants.
 - Contaminants: debris, sugar residue, acids, other impurities, etc.
- PC5.**
 - maintain alignment and ensure proper assembly of parts.
 - Parts: loose bolts, nuts, fasteners, etc.
- PC6.** replace filters and screens per maintenance plan recommendations in order to avoid clogging
- PC7.** examine or replace drive systems, chains, and conveyor belts to avoid belt or chain failures
- PC8.**
 - inspect appropriate parts for loose connections or corrosion.
 - Parts: wiring, control panels, electrical connections, etc.
- PC9.** verify sensors, switches and control systems are operating properly
- PC10.**
 - monitor cooling and heating systems for technical issues
 - Cooling and Heating system: heat exchangers, condensers, etc.
 - Technical issues: leaks, obstructions, temperature changes, etc.
- PC11.** examine or replace damaged seals, gaskets and lubricate pumps if required
- PC12.**
 - make sure safety precautions are working as intended
 - Safety precautions: emergency stops, safety guards, etc.
- PC13.** adhere to preventive maintenance schedule created by maintenance department

Perform troubleshooting of equipment issues

To be competent, the user/individual on the job must be able to:

- PC14.** safely shutdown or isolate damaged equipment to avoid further harm

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- PC15.** • identify equipment problems that can assist in comprehending source of issue
• Problem identification: through strange sensations, noises, visual indications, etc.
- PC16.** conduct methodical testing and isolation of components or subsystems to identify precise location and cause of problem
- PC17.** check wiring, connections and device parts for signs of wear, damage and loose connections
- PC18.** • inquire about equipment operating circumstances
• Circumstances: current maintenance history, operating parameters, modifications to process conditions, etc.
- PC19.** • evaluate performance of equipment and collect information by using diagnostic tools and equipment
• Tools and equipment: meters, gauges, sensors, etc.
- PC20.** • interpret instrument readings and use it for troubleshooting along with coworkers
• Coworkers: managers, engineers, maintenance staff, etc.
- PC21.** categorize any possible contributing elements or circumstances that may result in recurrent issues and suggest preventive steps to avert such issues

Carry out documentation to ensure compliance

To be competent, the user/individual on the job must be able to:

- PC22.** • gather information and keep records of each manufacturing batch's information
• Information: raw material consumption, production rates, processing characteristics, equipment used, people involved, start and end time, etc.
- PC23.** create daily or shift production logs to track sugar processing status
- PC24.** record correctly and legibly the findings of quality testing in quality control documents
- PC25.** • record sampling process followed in production line
• sampling record: includes sample sizes, sampling locations, sample collecting techniques, etc.
- PC26.** • maintain records of normal maintenance procedures
• Procedures: repairs, calibrations, inspections, etc.
- PC27.** • ascertain all records are maintained per industry-specific regulations
• Regulations: laws governing food safety, environment, product quality, etc.
- PC28.** • discuss documentation findings with appropriate parties to address concerns or issues
• Parties: management, quality control teams, supervisors, etc.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** functions, components and operation of the processing equipment such as crushers, centrifuges, evaporators, packaging machinery, etc.
- KU2.** preventive maintenance procedures such as maintenance intervals, lubrication requirements, inspection points, etc.
- KU3.** how to use tools and equipment such as wrenches, lubrication devices, diagnostic instruments, etc.
- KU4.** safety protocols and best practices for maintenance tasks
- KU5.** importance of maintenance schedules and adherence
- KU6.** working of critical components and processes
- KU7.** data instruments and sensors

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- KU8.** how to determine root cause of the issues and which solution to implement
- KU9.** specific industry regulations, quality standards, and compliance requirements including food safety and environmental regulations
- KU10.** how to maintain accurate production, quality equipment maintenance records

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** focus on details while inspecting equipment and follow maintenance checklists
- GS2.** knowledge of documentation and record-keeping
- GS3.** analyze information and take appropriate actions
- GS4.** prioritize safety and adhering to safety protocols
- GS5.** note down the information communicated by the senior/supervisor, observations (if any) related to the process, as per applicability in the organization
- GS6.** read and interpret the work order from the supervisor, certificate of analysis, analysis report from internal lab, process required for producing various types of products, equipment manuals and process documents to understand the equipment operation and process requirement
- GS7.** communicate effectively with subordinates as well as supervisors
- GS8.** perform calculations related to quantities, parameters and measurements
- GS9.** collaborate and contribute to a positive and productive work environment
- GS10.** discuss task lists, schedules, and activities for carrying out production with the senior/supervisor

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Carry out preventive planned maintenance</i>	13	20	-	13
PC1. - check for wear, damage, or malfunction on a regular basis on all processing machinery. - Machinery: mills, juice heaters,clarifier, evaporators,pans, crystallisers, centrifuge, , drying units, etc.	-	-	-	-
PC2. perform operational and visual inspections to identify any anomalies or possible problems	-	-	-	-
PC3. lubricate appropriately moving components to decrease friction and prevent wear	-	-	-	-
PC4. - prevent performance issues by keeping equipment away from contaminants. - Contaminants: debris, sugar residue, acids, other impurities, etc.	-	-	-	-
PC5. - maintain alignment and ensure proper assembly of parts. - Parts: loose bolts, nuts, fasteners, etc.	-	-	-	-
PC6. replace filters and screens per maintenance plan recommendations in order to avoid clogging	-	-	-	-
PC7. examine or replace drive systems, chains, and conveyor belts to avoid belt or chain failures	-	-	-	-
PC8. - inspect appropriate parts for loose connections or corrosion. - Parts: wiring, control panels, electrical connections, etc.	-	-	-	-
PC9. verify sensors, switches and control systems are operating properly	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. - monitor cooling and heating systems for technical issues - Cooling and Heating system: heat exchangers, condensers, etc. - Technical issues: leaks, obstructions, temperature changes, etc.	-	-	-	-
PC11. examine or replace damaged seals, gaskets and lubricate pumps if required	-	-	-	-
PC12. - make sure safety precautions are working as intended - Safety precautions: emergency stops, safety guards, etc.	-	-	-	-
PC13. adhere to preventive maintenance schedule created by maintenance department	-	-	-	-
<i>Perform troubleshooting of equipment issues</i>	8	12	-	8
PC14. safely shutdown or isolate damaged equipment to avoid further harm	-	-	-	-
PC15. - identify equipment problems that can assist in comprehending source of issue - Problem identification: through strange sensations, noises, visual indications, etc.	-	-	-	-
PC16. conduct methodical testing and isolation of components or subsystems to identify precise location and cause of problem	-	-	-	-
PC17. check wiring, connections and device parts for signs of wear, damage and loose connections	-	-	-	-
PC18. - inquire about equipment operating circumstances - Circumstances: current maintenance history, operating parameters, modifications to process conditions, etc.	-	-	-	-
PC19. - evaluate performance of equipment and collect information by using diagnostic tools and equipment - Tools and equipment: meters, gauges, sensors, etc.	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC20. - interpret instrument readings and use it for troubleshooting along with coworkers - Coworkers: managers, engineers, maintenance staff, etc.	-	-	-	-
PC21. categorize any possible contributing elements or circumstances that may result in recurrent issues and suggest preventive steps to avert such issues	-	-	-	-
<i>Carry out documentation to ensure compliance</i>	7	12	-	7
PC22. - gather information and keep records of each manufacturing batch's information - Information: raw material consumption, production rates, processing characteristics, equipment used, people involved, start and end time, etc.	-	-	-	-
PC23. create daily or shift production logs to track sugar processing status	-	-	-	-
PC24. record correctly and legibly the findings of quality testing in quality control documents	-	-	-	-
PC25. - record sampling process followed in production line - sampling record: includes sample sizes, sampling locations, sample collecting techniques, etc.	-	-	-	-
PC26. - maintain records of normal maintenance procedures - Procedures: repairs, calibrations, inspections, etc.	-	-	-	-
PC27. - ascertain all records are maintained per industry-specific regulations - Regulations: laws governing food safety, environment, product quality, etc.	-	-	-	-
PC28. - discuss documentation findings with appropriate parties to address concerns or issues - Parties: management, quality control teams, supervisors, etc.	-	-	-	-
NOS Total	28	44	-	28

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	FIC/N7104
NOS Name	Carry out Maintenance and Troubleshooting
Sector	Food Processing
Sub-Sector	
Occupation	Processing-Confectionery
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	29/05/2027
NSQC Clearance Date	30/05/2024

Qualification Pack

FIC/N9906: Apply food safety guidelines in Food Processing

Description

This unit covers the essential components of food safety, Good Manufacturing Practices (GMP), and personal hygiene in the food industry. It emphasizes the importance of individuals working in the food industry in protecting the health and well-being of consumers by following food safety protocols and procedures and ensuring the production of safe and high-quality food products.

Scope

The scope covers the following :

- Apply personal hygiene and follow Good Manufacturing practices at the workplace.
- Implement Food Safety and pre-requisite programs (PRP) at the workplace.

Elements and Performance Criteria

Apply personal hygiene and follow Good Manufacturing practices at workplace

To be competent, the user/individual on the job must be able to:

- PC1.** PC1. follow a site relevant documented procedure for Personal Hygiene and Visitor/ Contractor rules.
- PC2.** PC2. follow work instructions at levels of employees inside a food manufacturing site and ensure that the relevant instructions are well communicated and being followed at the fixed timelines.
- PC3.** PC3. ensure timely participate and carry out the relevant training and awareness sessions on personal hygiene, GMP, and related topics.
- PC4.** PC4. ensure timely medical examination from a prescribed and authorized doctor and comply with the guidelines of Schedule IV as described in Food Safety Standard Authority of India (FSSAI) guidelines.
- PC5.** PC5. fill in data in the daily monitoring checklist related to personal hygiene, food safety, and GMP.
- PC6.**
 - PC6. follow a site-relevant documented procedure and area-wise work instructions for Good Manufacturing Practices (GMP) to be followed on the site.
 - procedure: Hand washing requirements, Gowning & De gowning protocols, cleaning, and sanitation of employee lockers, follow the protocols as laid down in the different categories of processing areas like Low Risk, High Risk, High Care areas, etc.
- PC7.** PC7. follow all validated Do's & Don'ts inside a food manufacturing firm.
- PC8.** PC8. follow man and materials movement throughout the production facility, to restrict unwanted hazards to cross-contaminate the products which are being manufactured in the facility.
- PC9.** PC9. refer to the process flow charts, HACCP summary plan, and critical process parameters in each and respective areas of the production line.
- PC10.** PC10. identify the material requirements such as manufacturing equipments, Utensils, and other processing aids, cleaning chemicals, and cleaning work instructions in all the relevant areas of the manufacturing facility. Also, a special focus shall be given to Allergens and their risks. Wherever required, the allergen requirements shall be separately addressed.

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- PC11.** PC11. ensure to properly tag and number all the equipment, machinery, tools, and other processing aids to keep proper traceability of the product being manufactured and handled at the site.
- PC12.** PC12. follow and implement all training and awareness guidelines in the manufacturing area and regularly participate in training effectiveness for evaluation.
- PC13.** PC13. participate in audits and address the aspects of Good Manufacturing Procedures, personal hygiene, and food safety.
- PC14.** PC14. ensure the record keeping and documentation such as Daily Monitoring Sheets, Batch Traceability Records, machine records, product parameters, process control parameters, etc.

Implement food safety practices at the workplace

To be competent, the user/individual on the job must be able to:

- PC15.** PC15. maintain updated facilities, equipment, and tool and design requirements to minimize the risks associated with the products being handled at the site.
- PC16.** PC16. follow the instruction in the raw and packaging materials warehouse and ensure receiving material parameters match all the laid requirements. parameters: Incoming vehicles Visual report, storage, and handling requirements, hazardous and non-hazardous goods, allergens, cross-contamination risks, Quarantine, Accepted & rejected goods, monitoring temperature and humidity, etc.
- PC17.** PC17. follow FSSAI Schedule IV requirements related to Pest Control, Cleaning, and Sanitation, Utilities, Waste Disposal, Prevention of Cross-Contamination, allergen management, corrective action, preventive actions, food operation control etc.
- PC18.** PC18. ensure timely check of the critical control points and product parameters.
- PC19.** PC19. record keeping and documentation such as daily monitoring sheets, cleaning sheets, parameters, etc.
- PC20.** PC20. report any food safety and GMP issue to the supervisor, if any.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** importance of personal hygiene, GMP, visitors & contractor's rules. Associated risk in case of deviation from the standard policies and how the requirement is linked with the site's FSSAI License.
- KU2.** KU2. importance of training and work instruction delivered by the supervisors.
- KU3.** KU3. importance of filling the records and checklists, formats and how to ensure that the timely and effective completion is achieved.
- KU4.** KU4. knowledge of trainings and skills required to perform in food processing premises.
- KU5.** KU5. understand FSSAI Schedule IV requirements of food handlers and PRPs within the processing area
- KU6.** KU6. importance of timely medical examinations and awareness of communicable diseases
- KU7.** Understanding of Do's & Don'ts, intellect mindset to understand the visual illustrations
- KU8.** KU8. understanding about Site Zoning plans.
- KU9.** KU9. awareness of layout which would help to demarcate the defined movements of RM, PM, FG, and wastes generated during the processing of goods. This one lays a framework to launch Good Manufacturing Practices (GMP) successfully and effectively on site.

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- KU10.** KU10. understand the manufacturing process, product parameters and process control parameters such as CCPs
- KU11.** KU11. understanding about Hazard Analysis and Critical Control Points (HACCP)
- KU12.** KU12. understanding about Allergens and their types and controls to monitor effective handling of allergen raw materials on site.
- KU13.** KU13. basic understanding of traceability and mock recall
- KU14.** KU14. awareness about Internal & external Audits
- KU15.** KU15. understanding for RCA CAPA, cleaning and sanitation
- KU16.** KU16. awareness about record keeping and data monitoring in various sheets as per organizational requirement

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** GS1. read and comprehend basic content to read labels, charts, signages, symbols and product manuals
- GS2.** GS2. communicate with coworkers appropriately to clarify instructions and other issues
- GS3.** GS3. plan and organize the work schedule, work area, tools, equipment, and materials for improved productivity
- GS4.** GS4. plan and prioritize tasks as per work requirements
- GS5.** GS5. always be punctual and courteous
- GS6.** GS6. good observations and intellect mindset

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Apply personal hygiene and follow Good Manufacturing practices at workplace</i>	22	44	-	6
PC1. PC1. follow a site relevant documented procedure for Personal Hygiene and Visitor/ Contractor rules.	2	4	-	-
PC2. PC2. follow work instructions at levels of employees inside a food manufacturing site and ensure that the relevant instructions are well communicated and being followed at the fixed timelines.	2	4	-	2
PC3. PC3. ensure timely participate and carry out the relevant training and awareness sessions on personal hygiene, GMP, and related topics.	2	4	-	-
PC4. PC4. ensure timely medical examination from a prescribed and authorized doctor and comply with the guidelines of Schedule IV as described in Food Safety Standard Authority of India (FSSAI) guidelines.	2	4	-	-
PC5. PC5. fill in data in the daily monitoring checklist related to personal hygiene, food safety, and GMP.	2	4	-	-
PC6. - PC6. follow a site-relevant documented procedure and area-wise work instructions for Good Manufacturing Practices (GMP) to be followed on the site. - procedure: Hand washing requirements, Gowning & De gowning protocols, cleaning, and sanitation of employee lockers, follow the protocols as laid down in the different categories of processing areas like Low Risk, High Risk, High Care areas, etc.	2	4	-	2
PC7. PC7. follow all validated Do's & Don'ts inside a food manufacturing firm.	1	2	-	1
PC8. PC8. follow man and materials movement throughout the production facility, to restrict unwanted hazards to cross-contaminate the products which are being manufactured in the facility.	2	4	-	-
PC9. PC9. refer to the process flow charts, HACCP summary plan, and critical process parameters in each and respective areas of the production line.	1	2	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. PC10. identify the material requirements such as manufacturing equipments, Utensils, and other processing aids, cleaning chemicals, and cleaning work instructions in all the relevant areas of the manufacturing facility. Also, a special focus shall be given to Allergens and their risks. Wherever required, the allergen requirements shall be separately addressed.	2	4	-	-
PC11. PC11. ensure to properly tag and number all the equipment, machinery, tools, and other processing aids to keep proper traceability of the product being manufactured and handled at the site.	1	2	-	-
PC12. PC12. follow and implement all training and awareness guidelines in the manufacturing area and regularly participate in training effectiveness for evaluation.	1	2	-	-
PC13. PC13. participate in audits and address the aspects of Good Manufacturing Procedures, personal hygiene, and food safety.	1	2	-	-
PC14. PC14. ensure the record keeping and documentation such as Daily Monitoring Sheets, Batch Traceability Records, machine records, product parameters, process control parameters, etc.	1	2	-	-
<i>Implement food safety practices at the workplace</i>	8	16	-	4
PC15. PC15. maintain updated facilities, equipment, and tool and design requirements to minimize the risks associated with the products being handled at the site.	2	4	-	-
PC16. PC16. follow the instruction in the raw and packaging materials warehouse and ensure receiving material parameters match all the laid requirements. parameters: Incoming vehicles Visual report, storage, and handling requirements, hazardous and non-hazardous goods, allergens, cross-contamination risks, Quarantine, Accepted & rejected goods, monitoring temperature and humidity, etc.	1	2	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC17. PC17. follow FSSAI Schedule IV requirements related to Pest Control, Cleaning, and Sanitation, Utilities, Waste Disposal, Prevention of Cross-Contamination, allergen management, corrective action, preventive actions, food operation control etc.	2	4	-	2
PC18. PC18. ensure timely check of the critical control points and product parameters.	1	2	-	-
PC19. PC19. record keeping and documentation such as daily monitoring sheets, cleaning sheets, parameters, etc.	1	2	-	1
PC20. PC20. report any food safety and GMP issue to the supervisor, if any.	1	2	-	-
NOS Total	30	60	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	FIC/N9906
NOS Name	Apply food safety guidelines in Food Processing
Sector	Food Processing
Sub-Sector	Generic
Occupation	Generic
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2029
NSQC Clearance Date	06/02/2026

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

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Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

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- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2029
NSQC Clearance Date	06/02/2026

Qualification Pack

FIC/N7105: Perform clarification of sugarcane juice in a sugar mill

Description

This OS unit is about removing the impurities from the sugarcane juice obtained from the milling process as per the industry standard and the available machinery.

Scope

The scope covers the following :

- Prepare tools for clarification process at a sugar mill
- Carryout clarification of sugarcane juice
- Perform post-production cleaning and regular maintenance of equipment

Elements and Performance Criteria

Prepare tools for clarification process at a sugar mill

To be competent, the user/individual on the job must be able to:

- PC1.** check the availability and working of the tools in the clarification area
- PC2.** inspect the workers in the station are wearing PPE kits
- PC3.** ensure all safety procedures are followed while working with high-temperature drying equipment

Carry out clarification of sugarcane juice

To be competent, the user/individual on the job must be able to:

- PC4.** pass the sugarcane juice to the vibrating screens/sieve baskets to remove solid particles, leaves and large debris
- PC5.** add prescribed amount of Sulphur dioxide or its derivatives Sulphur metabisulphite to maintain the color and prolong the shelf life
- PC6.** heat the raw sugarcane juice at 70-75° C temperature in primary heaters to kill the bacteria (Leuconostoc)
- PC7.** transfer the raw juice to the lime station to add lime (calcium hydroxide) to the solution to aid precipitation and coagulation of particles, neutralize the pH level
- PC8.** pump the neutralized cane juice into continuous clarifier and add flocculants (to improve sedimentation or filterability of smaller particles)
- PC9.** allow the treated juice to settle in settling tanks, which gives sludge at the bottom and clear juice on the top
- PC10.** clear juice is skimmed or decanted from the top and the sludge/mud is disposed of for use as fertilizers
- PC11.** transfer the clear juice to vacuum evaporators for further processing
- PC12.** document the activities of the clarification process for quality audit and process improvement

Perform post-production cleaning and regular maintenance of equipment

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC13.** clean the clarification process equipment with the approved cleaning agents and methods of cleaning
- PC14.** follow the GMP health and safety standards while cleaning equipment
- PC15.** sanitize the work with the approved sanitizers and as per GMP standards
- PC16.** handle the maintenance needs/repairs of equipment that are identified during cleaning activity
- PC17.** document the equipment maintenance and cleaning activities
- PC18.** dispose of waste materials gathered during the cleaning process

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** benefits of heating the raw sugar cane juice
- KU2.** importance of clarification process in sugar cane processing
- KU3.** role of Sulphur-di-oxide in sugar processing
- KU4.** reasons for using high-quality milk of lime
- KU5.** importance of the pH level
- KU6.** flocculation process and flocculants used
- KU7.** advantages of flocculation process
- KU8.** skimming or decanting of clarified sugar cane juice
- KU9.** working of the various tools used in the clarification process
- KU10.** chemical composition of raw sugar cane juice and clarified juice
- KU11.** attributes to check as part of Quality control
- KU12.** waste disposal as per the industry standards and GMP guidelines
- KU13.** Industry best practices for manufacturing goods (GMP)
- KU14.** FSSAI schedule 4 standards with regard to manufacturing food products
- KU15.** personal health and safety standards while at work

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret policies and process manuals
- GS2.** communicate with others effectively
- GS3.** collaborate with team members to increase process efficiency
- GS4.** plan and prioritize tasks to maximize productivity
- GS5.** be punctual and courteous
- GS6.** adhere to the production schedule and meet deadline
- GS7.** follow safety standards for self and others
- GS8.**
 - read operation manuals and process documents to understand the equipment operation and process
 - requirement

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare tools for clarification process at a sugar mill</i>	5	6	-	3
PC1. check the availability and working of the tools in the clarification area	1	2	-	1
PC2. inspect the workers in the station are wearing PPE kits	2	2	-	1
PC3. ensure all safety procedures are followed while working with high-temperature drying equipment	2	2	-	1
<i>Carry out clarification of sugarcane juice</i>	12	21	-	13
PC4. pass the sugarcane juice to the vibrating screens/sieve baskets to remove solid particles, leaves and large debris	2	3	-	2
PC5. add prescribed amount of Sulphur dioxide or its derivatives Sulphur metabisulphite to maintain the color and prolong the shelf life	2	3	-	2
PC6. heat the raw sugarcane juice at 70-75° C temperature in primary heaters to kill the bacteria (Leuconostoc)	1	2	-	1
PC7. transfer the raw juice to the lime station to add lime (calcium hydroxide) to the solution to aid precipitation and coagulation of particles, neutralize the pH level	1	2	-	1
PC8. pump the neutralized cane juice into continuous clarifier and add flocculants (to improve sedimentation or filterability of smaller particles	1	2	-	1
PC9. allow the treated juice to settle in settling tanks, which gives sludge at the bottom and clear juice on the top	1	2	-	1
PC10. clear juice is skimmed or decanted from the top and the sludge/mud is disposed of for use as fertilizers	2	3	-	2

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. transfer the clear juice to vacuum evaporators for further processing	1	2	-	1
PC12. document the activities of the clarification process for quality audit and process improvement	1	2	-	2
<i>Perform post-production cleaning and regular maintenance of equipment</i>	12	18	-	10
PC13. clean the clarification process equipment with the approved cleaning agents and methods of cleaning	2	3	-	2
PC14. follow the GMP health and safety standards while cleaning equipment	2	4	-	1
PC15. sanitize the work with the approved sanitizers and as per GMP standards	2	2	-	2
PC16. handle the maintenance needs/repairs of equipment that are identified during cleaning activity	2	4	-	1
PC17. document the equipment maintenance and cleaning activities	2	2	-	2
PC18. dispose of waste materials gathered during the cleaning process	2	3	-	2
NOS Total	29	45	-	26

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	FIC/N7105
NOS Name	Perform clarification of sugarcane juice in a sugar mill
Sector	Food Processing
Sub-Sector	
Occupation	Processing-Confectionery
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	29/05/2027
NSQC Clearance Date	30/05/2024

Qualification Pack

FIC/N7106: Perform Evaporation of sugarcane Juice in a sugar mill

Description

This OS unit is about overseeing the evaporation stage of sugar production in order to get a concentrated sugarcane juice for extracting sugar crystals as per the industry standard and the available machinery.

Scope

The scope covers the following :

- The scope covers the following
- Prepare the tools for evaporation process
- Carryout evaporation of sugarcane juice at the sugar mill
- Perform post-production cleaning and regular maintenance of equipment

Elements and Performance Criteria

Prepare the tools for evaporation process

To be competent, the user/individual on the job must be able to:

- PC1.** check the evaporators for the process are working efficiently and safety
- PC2.** check the quality of chemicals and raw materials used for the process are available in adequate quantity
- PC3.** ensure all safety procedures are followed while working with high-temperature drying equipment
- PC4.** check the availability of health and safety kits in the work area

Carryout evaporation of sugarcane juice at the sugar mill

To be competent, the user/individual on the job must be able to:

- PC5.**
 - adjust various parameters of the evaporators to get the desired concentration of sugar cane juice
 - Parameters: temperature, pressure, flow rates
- PC6.** preheat the clarified sugarcane solution to improve the energy efficiency of the evaporation process
- PC7.**
 - add appropriate defoaming agents to prevent foam formation
 - defoaming agents' type: silicone-based, polyglycol-based, mineral oil-based, fatty acid-based, non-ionic surfactants, food grade antifoam agents
- PC8.** transfer this solution to the series of multiple-effect evaporators in a controlled temperature
- PC9.** pass steam through the first evaporator which vaporizes the water in the sugarcane juice
- PC10.** vapour from the first evaporator moves to the second evaporator in the series and works as a heating medium
- PC11.** ensure the sugarcane juice flows through multiple effect evaporators to get a thicker syrup at each step
- PC12.** check the concentration level of the syrup at the end of the process, is as per the industry standards (from 10-15 Brix dissolved solids to 50-70 Brix thick syrup)
- PC13.** transfer the concentrated syrup to cooling chambers to reduce their temperature

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- PC14.** • send sample to the appropriate department for quality check
• quality check: sugar content, purity
- PC15.** store them in tanks or vessels until ready for the next process namely crystallization
- PC16.** document the activities of the evaporation process for quality audits and process improvement

perform post-production cleaning and regular maintenance of equipment

To be competent, the user/individual on the job must be able to:

- PC17.** clean the equipment with the approved cleaning agents and methods of cleaning
- PC18.** follow the GMP health and safety standards while cleaning equipment
- PC19.** sanitize the work area with the approved sanitizers and as per GMP standards
- PC20.** check the working of the equipment before the next batch is ready for evaporation process
- PC21.** document the cleaning and maintenance activities
- PC22.** dispose of waste materials gathered during the cleaning process

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** importance of preheating the sugarcane juice/extract
- KU2.** significance of the evaporation process
- KU3.** tools used in the evaporation process
- KU4.** working of the multi-effect evaporators
- KU5.** about defoaming agents
- KU6.** criteria for selection of defoaming agents and prescribed dosage
- KU7.** effects of foaming of sugarcane juice in the evaporation process
- KU8.** process of condensation
- KU9.** importance of the brix level of the concentrated syrup
- KU10.** importance of cooling the sugar syrup to reduce its temperature
- KU11.** check the sugar content and purity as part of Quality control
- KU12.** Industry Best Practices for manufacturing goods (GMP)
- KU13.** FSSAI schedule 4 standards with regard to manufacturing food products
- KU14.** personal health and safety standards while at work

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret policies and process manuals
- GS2.** communicate with others effectively
- GS3.** collaborate with team members to increase process efficiency
- GS4.** plan and prioritize tasks to maximize productivity
- GS5.** be punctual and courteous

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- GS6.** adhere to the production schedule and meet deadline
- GS7.** follow safety standards for self and others
- GS8.** read operation manuals and process documents to understand the equipment operation and process requirement

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare the tools for evaporation process</i>	5	8	-	4
PC1. check the evaporators for the process are working efficiently and safety	1	2	-	1
PC2. check the quality of chemicals and raw materials used for the process are available in adequate quantity	2	2	-	1
PC3. ensure all safety procedures are followed while working with high-temperature drying equipment	1	2	-	1
PC4. check the availability of health and safety kits in the work area	1	2	-	1
<i>Carryout evaporation of sugarcane juice at the sugar mill</i>	12	23	-	11
PC5. • adjust various parameters of the evaporators to get the desired concentration of sugar cane juice • Parameters: temperature, pressure, flow rates	1	2	-	1
PC6. preheat the clarified sugarcane solution to improve the energy efficiency of the evaporation process	1	3	-	1
PC7. • add appropriate defoaming agents to prevent foam formation • defoaming agents' type: silicone-based, polyglycol-based, mineral oil-based, fatty acid-based, non-ionic surfactants, food grade antifoam agents	1	1	-	1
PC8. transfer this solution to the series of multiple-effect evaporators in a controlled temperature	1	1	-	1
PC9. pass steam through the first evaporator which vaporizes the water in the sugarcane juice	1	2	-	1
PC10. vapour from the first evaporator moves to the second evaporator in the series and works as a heating medium	1	2	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. ensure the sugarcane juice flows through multiple effect evaporators to get a thicker syrup at each step	1	2	-	-
PC12. check the concentration level of the syrup at the end of the process, is as per the industry standards (from 10-15 Brix dissolved solids to 50-70 Brix thick syrup)	1	2	-	1
PC13. transfer the concentrated syrup to cooling chambers to reduce their temperature	1	2	-	1
PC14. • send sample to the appropriate department for quality check • quality check: sugar content, purity	1	2	-	1
PC15. store them in tanks or vessels until ready for the next process namely crystallization	1	2	-	1
PC16. document the activities of the evaporation process for quality audits and process improvement	1	2	-	1
<i>perform post-production cleaning and regular maintenance of equipment</i>	11	17	-	9
PC17. clean the equipment with the approved cleaning agents and methods of cleaning	1	2	-	1
PC18. follow the GMP health and safety standards while cleaning equipment	2	4	-	1
PC19. sanitize the work area with the approved sanitizers and as per GMP standards	2	2	-	2
PC20. check the working of the equipment before the next batch is ready for evaporation process	2	4	-	1
PC21. document the cleaning and maintenance activities	2	2	-	2
PC22. dispose of waste materials gathered during the cleaning process	2	3	-	2
NOS Total	28	48	-	24

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	FIC/N7106
NOS Name	Perform Evaporation of sugarcane Juice in a sugar mill
Sector	Food Processing
Sub-Sector	
Occupation	Processing-Confectionery
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	29/05/2027
NSQC Clearance Date	30/05/2024

Qualification Pack

FIC/N7107: Perform crystallization of sugarcane Juice in a sugar mill

Description

This OS unit is about overseeing the crystallization process in which the concentrated sugar syrup is converted into sugar crystals as per the industry standard and the available machineries.

Scope

The scope covers the following :

- The scope covers the following
- Prepare the tools for crystallization process
- Carryout crystallization of concentrated sugar syrup at the sugar mill
- Perform post production cleaning and regular maintenance of equipment

Elements and Performance Criteria

Prepare the tools for crystallization process

To be competent, the user/individual on the job must be able to:

- PC1.** verify the crystallizers (large vessel with agitators or mixers) used for the crystallization process are working efficiently and safety
- PC2.** check the chemicals and raw materials used for the process are available in adequate quantity
- PC3.** verify the work area is clean and is ready for the crystallization process
- PC4.** check the availability of health and safety kits at the work area
- PC5.** ensure all safety procedures are followed while working with high temperature drying equipment

Carryout crystallization of concentrated sugar syrup at the sugar mill

To be competent, the user/individual on the job must be able to:

- PC6.**
 - adjust various parameters of the vacuum pan to get the desired sugar crystals
 - Parameters: temperature, pressure, vacuum levels
- PC7.** feed the sugar syrup into the vacuum pan crystallizers to begin the crystallization process
- PC8.** add small quantity of sugar crystals or seed crystals to initiate the crystallization process
- PC9.** monitor the seeding process for uniform crystal formation
- PC10.** check the seed crystals are distributed evenly using agitators/stirrers
- PC11.**
 - control supersaturation with these measures
 - Measures: seed crystals, agitator speed, antisolvents or crystal growth modifiers, pH control,
- PC12.**
 - monitor and test the progress of crystallization
 - Parameters: sugar content, crystal size, quality of the growing sugar crystals
- PC13.** control the cooling rate to promote uniform growth of sugar crystals
- PC14.** leave the mass of sugar crystals in the crystallizers after draining the syrup namely the molasses
- PC15.** transfer the sugar crystals to centrifuge separate sugar crystals from the remaining syrup (centrifuge rotates at 1000-2800 revolutions per minute)

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- PC16.** use spring water to wash the crystals off the centrifuge
- PC17.** • send sample to the appropriate department for quality check
• Quality check: flavor, color, purity, size
- PC18.** send the sugar crystals for drying station for drying process
perform post-production cleaning and regular maintenance of equipment
- To be competent, the user/individual on the job must be able to:
- PC19.** clean the equipment with the approved cleaning agents and methods of cleaning
- PC20.** disassembling and cleaning separator components to prevent residual sugar accumulation and contamination
- PC21.** maintain the dust collection systems and filters to prevent air borne dust that can contaminate the sugar
- PC22.** follow the GMP health and safety standards while cleaning equipment
- PC23.** sanitize the work area with the approved sanitizers and as per GMP standards
- PC24.** handle the maintenance needs/repairs of equipment that are identified during cleaning activity
- PC25.** document the equipment maintenance and cleaning activities
- PC26.** dispose of waste materials gathered during the cleaning process

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** crystallization process in sugar production
- KU2.** working of the vacuum pan crystallizers and other tools used in this process
- KU3.** significance of adding seed crystals
- KU4.** need for monitoring growth of sugar crystals
- KU5.** factors that influence the crystallization of sugar
- KU6.** massecuite and mother liquor
- KU7.** centrifugation process to separate the crystals from the mother liquor
- KU8.** supersaturation process
- KU9.** significance of monitoring the cooling process
- KU10.** byproduct molasses
- KU11.** check the sugar flavor, color, purity and size as part of Quality control
- KU12.** documentation of crystallization process, cleaning and maintenance activities
- KU13.** industry best practices for manufacturing goods (GMP)
- KU14.** FSSAI schedule 4 standards with regard to manufacturing food products
- KU15.** personal health and safety standards while at work

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret policies and process manuals

Qualification Pack

- GS2.** communicate with others effectively
- GS3.** collaborate with team members to increase process efficiency
- GS4.** plan and prioritize tasks to maximize productivity
- GS5.** be punctual and courteous
- GS6.** adhere to the production schedule and meet deadline
- GS7.** follow safety standards for self and others
- GS8.** read operation manuals and process documents to understand the equipment operation and process requirement

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare the tools for crystallization process</i>	5	10	-	5
PC1. verify the crystallizers (large vessel with agitators or mixers) used for the crystallization process are working efficiently and safety	1	2	-	1
PC2. check the chemicals and raw materials used for the process are available in adequate quantity	1	2	-	1
PC3. verify the work area is clean and is ready for the crystallization process	1	2	-	1
PC4. check the availability of health and safety kits at the work area	1	2	-	1
PC5. ensure all safety procedures are followed while working with high temperature drying equipment	1	2	-	1
<i>Carryout crystallization of concentrated sugar syrup at the sugar mill</i>	13	18	-	11
PC6. • adjust various parameters of the vacuum pan to get the desired sugar crystals • Parameters: temperature, pressure, vacuum levels	1	2	-	1
PC7. feed the sugar syrup into the vacuum pan crystallizers to begin the crystallization process	1	2	-	1
PC8. add small quantity of sugar crystals or seed crystals to initiate the crystallization process	1	2	-	1
PC9. monitor the seeding process for uniform crystal formation	1	1	-	1
PC10. check the seed crystals are distributed evenly using agitators/stirrers	1	1	-	1
PC11. • control supersaturation with these measures • Measures: seed crystals, agitator speed, antisolvents or crystal growth modifiers, pH control,	1	2	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. • monitor and test the progress of crystallization • Parameters: sugar content, crystal size, quality of the growing sugar crystals	1	2	-	1
PC13. control the cooling rate to promote uniform growth of sugar crystals	1	1	-	1
PC14. leave the mass of sugar crystals in the crystallizers after draining the syrup namely the molasses	1	1	-	1
PC15. transfer the sugar crystals to centrifuge separate sugar crystals from the remaining syrup (centrifuge rotates at 1000-2800 revolutions per minute)	1	1	-	1
PC16. use spring water to wash the crystals off the centrifuge	1	1	-	-
PC17. • send sample to the appropriate department for quality check • Quality check: flavor, color, purity, size	1	1	-	1
PC18. send the sugar crystals for drying station for drying process	1	1	-	-
<i>perform post-production cleaning and regular maintenance of equipment</i>	12	18	-	8
PC19. clean the equipment with the approved cleaning agents and methods of cleaning	1	1	-	1
PC20. disassembling and cleaning separator components to prevent residual sugar accumulation and contamination	1	2	-	1
PC21. maintain the dust collection systems and filters to prevent air borne dust that can contaminate the sugar	1	1	-	1
PC22. follow the GMP health and safety standards while cleaning equipment	2	3	-	1
PC23. sanitize the work area with the approved sanitizers and as per GMP standards	2	2	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC24. handle the maintenance needs/repairs of equipment that are identified during cleaning activity	1	4	-	1
PC25. document the equipment maintenance and cleaning activities	2	2	-	1
PC26. dispose of waste materials gathered during the cleaning process	2	3	-	1
NOS Total	30	46	-	24

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	FIC/N7107
NOS Name	Perform crystallization of sugarcane juice in a sugar mill
Sector	Food Processing
Sub-Sector	
Occupation	Processing-Confectionery
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	29/05/2027
NSQF Clearance Date	30/05/2024

Qualification Pack

FIC/N7108: Perform refining of crystal sugar

Description

This OS unit is about managing refining process of the sugar crystals as per the industry standard and the available machinery.

Scope

The scope covers the following :

- The scope covers the following
- Organize the raw materials for refining process
- Carryout refining of sugar from sugar crystals
- Perform post-production cleaning and regular maintenance of equipment

Elements and Performance Criteria

Organize the raw materials for refining process

To be competent, the user/individual on the job must be able to:

- PC1.** procure the sugar crystals for refining
- PC2.** clean the sugar crystals through clarification, evaporation, crystallization and drying
- PC3.** check the refining machinery are in working condition
- PC4.** verify the work area is clean and ready for the refining process
- PC5.** check the availability of health and safety kits in the work area

Carryout refining of sugar from sugar crystals

To be competent, the user/individual on the job must be able to:

- PC6.** pass the dried sugar crystals into different vibrating screens / Sievers with different mesh sizes for grading
- PC7.** collect the graded sugar in a container
- PC8.** transfer the remaining to the vibrating screens / Sievers again with a different mesh size to get different grades of sugar
- PC9.** send the sugar of different grades for quality check
- PC10.** conduct the quality of sugar to check if it meets the specifications

perform post-production cleaning and regular maintenance of equipment

To be competent, the user/individual on the job must be able to:

- PC11.** clean and sanitize the screens, sieves and other equipment for correct sizing and grading with the approved cleaning agents and methods of cleaning
- PC12.** follow the GMP health and safety standards while cleaning equipment
- PC13.** sanitize the work area with the approved sanitizers and as per GMP standards
- PC14.** address the repair/replacement needs of equipment that were identified during cleaning
- PC15.** inspect the equipment and infrastructure for wear and tears
- PC16.** dispose of waste materials in line with industry and environmental regulations

Qualification Pack

PC17. document the cleaning and maintenance activities for internal audit

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** refining process in sugar production
- KU2.** Instrumentation and control systems
- KU3.** different methods of sorting and grading sugar crystals
- KU4.** tools used in sorting and grading
- KU5.** operating techniques of the tools
- KU6.** check the sugar color, purity and size as part of Quality control
- KU7.** Industry Best Practices for manufacturing goods (GMP)
- KU8.** FSSAI schedule 4 standards with regard to manufacturing food products
- KU9.** personal health and safety standards while at work

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret policies and process manuals
- GS2.** communicate with others effectively
- GS3.** collaborate with team members to increase process efficiency
- GS4.** plan and prioritize tasks to maximize productivity
- GS5.** be punctual and courteous
- GS6.** adhere to the production schedule and meet deadline
- GS7.** follow safety standards for self and others
- GS8.** read operation manuals and process documents to understand the equipment operation and process requirement

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Organize the raw materials for refining process</i>	5	10	-	5
PC1. procure the sugar crystals for refining	1	2	-	1
PC2. clean the sugar crystals through clarification, evaporation, crystallization and drying	1	2	-	1
PC3. check the refining machinery are in working condition	1	2	-	1
PC4. verify the work area is clean and ready for the refining process	1	2	-	1
PC5. check the availability of health and safety kits in the work area	1	2	-	1
<i>Carryout refining of sugar from sugar crystals</i>	11	15	-	9
PC6. pass the dried sugar crystals into different vibrating screens / Sievers with different mesh sizes for grading	3	4	-	2
PC7. collect the graded sugar in a container	2	2	-	1
PC8. transfer the remaining to the vibrating screens / Sievers again with a different mesh size to get different grades of sugar	2	4	-	2
PC9. send the sugar of different grades for quality check	2	2	-	2
PC10. conduct the quality of sugar to check if it meets the specifications	2	3	-	2
<i>perform post-production cleaning and regular maintenance of equipment</i>	15	20	-	10
PC11. clean and sanitize the screens, sieves and other equipment for correct sizing and grading with the approved cleaning agents and methods of cleaning	3	3	-	1
PC12. follow the GMP health and safety standards while cleaning equipment	2	4	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. sanitize the work area with the approved sanitizers and as per GMP standards	2	2	-	1
PC14. address the repair/replacement needs of equipment that were identified during cleaning	2	4	-	2
PC15. inspect the equipment and infrastructure for wear and tears	2	2	-	1
PC16. dispose of waste materials in line with industry and environmental regulations	2	3	-	2
PC17. document the cleaning and maintenance activities for internal audit	2	2	-	2
NOS Total	31	45	-	24

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	FIC/N7108
NOS Name	Perform refining of crystal sugar
Sector	Food Processing
Sub-Sector	
Occupation	Processing-Confectionery
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	29/05/2027
NSQC Clearance Date	30/05/2024

Qualification Pack

FIC/N7109: Perform drying of crystal sugar

Description

This OS unit is about removing the moisture from the sugar crystals in the drying section and ensure the final product meets the quality standards. They are also responsible for the maintenance and troubleshooting of the equipment used in the process.

Scope

The scope covers the following :

- The scope covers the following
- Prepare the tools for drying the raw sugar
- Carryout drying process in a sugar mill
- Perform post-production cleaning and regular maintenance of equipment

Elements and Performance Criteria

Prepare the tools for refining the raw sugar crystals

To be competent, the user/individual on the job must be able to:

- PC1.** check the availability and the working of the drying equipment for the drying process
- PC2.** verify the work area is clean and ready for the drying process
- PC3.** check the availability of health and safety kits in the work area
- PC4.** ensure all safety procedures are followed while working with high-temperature drying equipment

Carryout drying of raw sugar crystals at the sugar mill

To be competent, the user/individual on the job must be able to:

- PC5.** operate and monitor the drying equipment for efficient and safe working
- PC6.** adjust the temperature and airflow of the drying equipment
- PC7.** operate the conveyor for carrying the sugar to the dryer
- PC8.** supervise the loading of sugar crystals into the dryer for moisture removal
- PC9.** PC9. set the drying time for the sugar crystals based on the moisture content while loading and the desired moisture level at the end (recommended level 0.02% 0.06% by weight)
- PC10.**
 - send the dried sugar for quality checks
 - checks: moisture content
- PC11.** maintain records of the process carried out for quality audits and process optimization
- PC12.** transport the dried sugar for packaging

perform post-production cleaning and regular maintenance of equipment

To be competent, the user/individual on the job must be able to:

- PC13.** clean the dryers to remove residual sugar particles with the approved cleaning agents and methods of cleaning
- PC14.** maintain the dust collection systems and filters to prevent air borne dust that can contaminate the sugar

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- PC15.** follow the GMP health and safety standards while cleaning equipment
- PC16.** sanitize the work area with the approved sanitizers and as per GMP standards
- PC17.** handle the maintenance needs/repairs of equipment that are identified during cleaning activity
- PC18.** document the equipment maintenance and cleaning activities

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** importance of drying process in sugar milling
- KU2.** working of the different types of driers used for removing moisture content in sugar
- KU3.** significance of maintaining the approved moisture level and ill effects of high levels of moisture
- KU4.** troubleshooting and maintenance of various tools used in this process
- KU5.** documentation of the drying process, cleaning and maintenance of tools
- KU6.** Industry Best Practices for Manufacturing Goods (GMP)
- KU7.** FSSAI schedule 4 standards with regard to manufacturing food products
- KU8.** personal health and safety standards while at work

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret policies and process manuals
- GS2.** communicate with others effectively
- GS3.** collaborate with team members to increase process efficiency
- GS4.** plan and prioritize tasks to maximize productivity
- GS5.** be punctual and courteous
- GS6.** adhere to the production schedule and meet deadline
- GS7.** follow safety standards for self and others
- GS8.** read operation manuals and process documents to understand the equipment operation and process requirement

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare the tools for refining the raw sugar crystals</i>	4	8	-	4
PC1. check the availability and the working of the drying equipment for the drying process	1	2	-	1
PC2. verify the work area is clean and ready for the drying process	1	2	-	1
PC3. check the availability of health and safety kits in the work area	1	2	-	1
PC4. ensure all safety procedures are followed while working with high-temperature drying equipment	1	2	-	1
<i>Carryout drying of raw sugar crystals at the sugar mill</i>	14	20	-	13
PC5. operate and monitor the drying equipment for efficient and safe working	2	4	-	2
PC6. adjust the temperature and airflow of the drying equipment	2	2	-	1
PC7. operate the conveyor for carrying the sugar to the dryer	1	2	-	1
PC8. supervise the loading of sugar crystals into the dryer for moisture removal	2	2	-	2
PC9. set the drying time for the sugar crystals based on the moisture content while loading and the desired moisture level at the end (recommended level 0.02% 0.06% by weight)	2	4	-	2
PC10. • send the dried sugar for quality checks • checks: moisture content	1	2	-	2
PC11. maintain records of the process carried out for quality audits and process optimization	2	2	-	1
PC12. transport the dried sugar for packaging	2	2	-	2
<i>perform post-production cleaning and regular maintenance of equipment</i>	12	15	-	10

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. clean the dryers to remove residual sugar particles with the approved cleaning agents and methods of cleaning	2	2	-	2
PC14. maintain the dust collection systems and filters to prevent air borne dust that can contaminate the sugar	2	2	-	2
PC15. follow the GMP health and safety standards while cleaning equipment	2	4	-	1
PC16. sanitize the work area with the approved sanitizers and as per GMP standards	2	2	-	2
PC17. handle the maintenance needs/repairs of equipment that are identified during cleaning activity	2	3	-	1
PC18. document the equipment maintenance and cleaning activities	2	2	-	2
NOS Total	30	43	-	27

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	FIC/N7109
NOS Name	Perform drying of crystal sugar
Sector	Food Processing
Sub-Sector	
Occupation	Processing-Confectionery
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	29/05/2027
NSQC Clearance Date	30/05/2024

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

This section includes the processes involved in identifying, gathering, and interpreting information to evaluate the Candidate on the required

competencies of the program.

<1. Assessment System Overview:

- ☐ Batches assigned to the assessment agencies for conducting the assessment on SIP or email
- ☐ Assessment agencies send the assessment confirmation to VTP/TC looping SSC
- ☐ Assessment agency deploys the ToA-certified Assessor to execute the assessment
- ☐ SSC monitors the assessment process & records

2. Testing Environment:

- ☐ Check the Assessment location, date and time

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- ☐ If the batch size is more than 30, then there should be 2 Assessors.
- ☐ Check that the allotted time to the candidates to complete the Theory & Practical Assessment is correct.

3. Assessment Quality Assurance levels/Framework:

- ☐ Question bank is created by the Subject Matter Experts (SME) are verified by the other SME
- ☐ Questions are mapped to the specified assessment criteria
- ☐ Assessor must be ToA certified & trainer must be ToT Certified

4. Types of evidence or evidence-gathering protocol:

- ☐ Time-stamped & geotagged reporting of the assessor from the assessment location
- ☐ Centre photographs with signboards and scheme-specific branding

5. Method of verification or validation:

- ☐ Surprise visit to the assessment location

6. Method for assessment documentation, archiving, and access

- ☐ Hard copies of the documents are stored

On the Job:

1. Each module (which covers the job profile of the Sugar Processing Operator) will be assessed separately.
2. The candidate must score 60% in each module to successfully complete the OJT.
3. Tools of Assessment that will be used for assessing whether the candidate is the desired skills and etiquette of dealing with customers,

understanding needs & requirements, assessing the customer and performing Soft Skills effectively:

a. Videos of Trainees during OJT

b.

4. Assessment of each Module will ensure that the candidate is able to:

a. Effective engagement with the customers

b. Understand the working of various tools and equipment....

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Qualification Pack

Minimum Passing % at NOS Level: 70

(Please note: A Trainee must score the minimum percentage for each NOS separately as well as on the QP as a whole.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
FIC/N9026.Prepare for production	30	70	-	-	100	10
FIC/N7103.Carry out sugar processing	23	45	-	32	100	30
FIC/N7104.Carry out Maintenance and Troubleshooting	28	44	-	28	100	20
FIC/N9906.Apply food safety guidelines in Food Processing	30	60	-	10	100	5
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	5
Total	131	249	-	70	450	70

Elective: 1 Clarification of sugarcane juice in a sugar mill

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
FIC/N7105.Perform clarification of sugarcane juice in a sugar mill	29	45	-	26	100	30
Total	29	45	-	26	100	30

Elective: 2 Perform evaporation of sugarcane Juice in a sugar mill

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
FIC/N7106.Perform Evaporation of sugarcane Juice in a sugar mill	28	48	-	24	100	30
Total	28	48	-	24	100	30

Elective: 3 Perform crystallization of sugar syrup in a sugar mill

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
FIC/N7107.Perform crystallization of sugarcane Juice in a sugar mill	30	46	-	24	100	30
Total	30	46	-	24	100	30

Elective: 4 Perform refining of crystal sugar

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
FIC/N7108.Perform refining of crystal sugar	31	45	-	24	100	30
Total	31	45	-	24	100	30

Elective: 5 Perform drying of crystal sugar-

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
FIC/N7109.Perform drying of crystal sugar	30	43	-	27	100	30

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
Total	30	43	-	27	100	30

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.